Pipefitter Field Guide

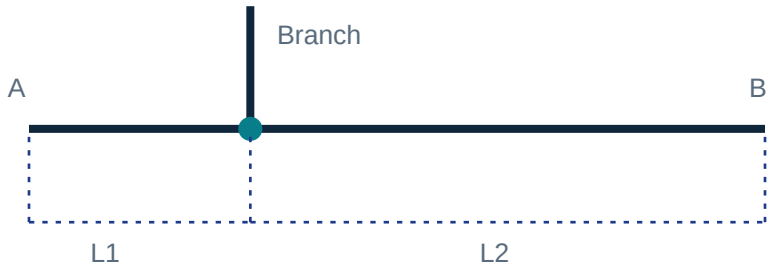
Pipefitter Field Guide: calculations, layouts, gas cutting, rigging and practical checks for the Pipefitter Field Tool app.

Use a datum, identify actual sizes, solve the geometry, account for fittings and verify against approved work documents.

Includes worked formulas, rigging calculations, model-specific torch guidance, wraparound marking and folded-paper saddle development. Example dimensions are for training.

Expanded October 2026 test edition. Examples explain methods; use the marked equipment and its current instructions for live work.

New practical layout section: straight reference lines with angle iron or strut, framing-square top marks, and eccentric reducer orientation (pages 64-69).



CONTENTS

Find it quickly

Choose a topic in the app reader, or use these page numbers in the PDF.

Topic	Pages
Measurements and layout geometry	3-22
ISO, spools and work checks	23-31
Technical references	32
Formulas and circumferences	33-34
Rigging and slings	35-39
Oxy-acetylene CUTTING charts	40-42
Wraparound and saddles	43-48
Original source references	49-50
Cutting and pipe OD/ID	51-53
Clearances and bolt circles	54-55
Flanges, reducers and slope	56-58
Cut lists, checks and tests	59-61
Field math and sources	62-63
Straight lines, top marks and reducers	64-69

START HERE

Use the right reference

Use this guide to organize measurements and check layout arithmetic. A sketch, a dimension and a fitting catalog each answer a different question. Keep their purposes separate.

The approved drawing defines the intended arrangement. The applicable specification and manufacturer documents identify acceptable components and installation requirements. This guide helps you connect those inputs without silently changing their meaning.

Examples use ideal geometry. Real pipe ends, fittings, joints, supports and access clearances still have to be accounted for. A correct triangle does not prove that a spool will fit.

This test edition is not a welding procedure, pressure-test procedure, engineering design or substitute for site training. Do not infer ratings, torque, wall thickness or allowable loading from its sketches.

FIELD CHECK Before relying on a result, identify the dimension basis, units, source revision and person responsible for approval.

FIELD NOTES / TEST FEEDBACK

Name both ends of a dimension

C-C means center-to-center. E-E means end-to-end. F-F means face-to-face. C-F and C-E connect different reference types. Write down which physical points each letter represents on your particular sketch.

A center may be the theoretical intersection of two pipe centerlines, not a point on the fitting surface. A face may be a flange sealing face; an end may be a prepared pipe end. These references are not interchangeable.

Do not enter a face-to-face measurement into a cut-length calculation expecting a center-to-center value. Convert using the actual component dimensions first.

Leader lines should terminate at the two intended reference points. Put dimension boxes away from crossings so another fitter can trace each value without guessing.



If A is a tee center and B is a flange sealing face, label the span C-F. Record the flange and tee takeoffs separately; do not call that span C-C.

MEASUREMENT

Fractions through sixteenths

Read the whole inches first, then the fraction. The table contains every positive sixteenth below one inch, reduced to its familiar form.

Converting 7/16 to decimal means dividing 7 by 16: 0.4375 in. A decimal such as 0.4375 is part of an inch, not part of a foot.

Fraction	Decimal in	Sixteenths
1/16	0.0625	1/16
1/8	0.1250	2/16
3/16	0.1875	3/16
1/4	0.2500	4/16
5/16	0.3125	5/16
3/8	0.3750	6/16
7/16	0.4375	7/16
1/2	0.5000	8/16
9/16	0.5625	9/16
5/8	0.6250	10/16
11/16	0.6875	11/16
3/4	0.7500	12/16
13/16	0.8125	13/16
7/8	0.8750	14/16
15/16	0.9375	15/16

FIELD CHECK A tape marked in sixteenths cannot justify reporting a measured length to a thousandth of an inch.

MEASUREMENT

Add and subtract cleanly

Put fractions over the same denominator before combining them. Keep the feet and inches visible so a carry or borrow is not lost.

For a long calculation, convert all lengths to inches, work at full precision, then convert the result back to feet and inches. Keep the original field measurement beside the converted value.

A negative answer in a cut-length calculation is a signal to recheck the input references or the available space. It is not a usable pipe length.

Addition: 2 ft 7-3/8 in + 1 ft 8-7/16 in.
Inches: 31-6/16 + 20-7/16 = 51-13/16.
Result: 4 ft 3-13/16 in.

Subtraction: 5 ft 2-1/4 in – 1 ft 9-5/8 in.
62-2/8 – 21-5/8 = 40-5/8 in.
Result: 3 ft 4-5/8 in.

FIELD CHECK Estimate first: about 31 + 20 is about 51 inches. A result near 5 inches or 500 inches needs correction.

FIELD NOTES / TEST FEEDBACK

MEASUREMENT

Round once, at the end

Choose the reporting increment before the final conversion. To round a nonnegative inch value to the nearest 1/16 in, multiply by 16, round to the nearest whole number, then divide by 16.

Keep full precision through intermediate trigonometry. Repeatedly rounding a travel, a takeoff and a remaining span can stack errors.

Rounding is a reporting decision, not a fabrication tolerance. The permitted deviation comes from the job requirements. At an exact halfway point, follow the project rounding convention consistently.

Calculated result: 17.386 in.
17.386 × 16 = 278.176.
Nearest whole sixteenth: 278.
278 ÷ 16 = 17.375 in = 17-3/8 in.
Rounding changed the value by 0.011 in.

FIELD CHECK Save the unrounded calculation in your notes when a later dimension depends on it.

FIELD NOTES / TEST FEEDBACK

MEASUREMENT

Keep units attached

One inch equals 25.4 millimeters exactly. One foot equals 12 inches or 304.8 millimeters. These are conversion definitions, not estimates. [1]

Convert a mixed feet-and-inches value to inches before multiplying by 25.4. Convert millimeters to inches by dividing by 25.4.

A unit conversion does not change a product designation into an equivalent purchasable size. Check the specified pipe or tube system rather than selecting a component solely from a converted nominal label.

3 ft 7-1/2 in = 43.5 in.
43.5 × 25.4 = 1104.9 mm.

For 850 mm: 850 ÷ 25.4 = 33.4646… in.
Nearest 1/16 in: 33-7/16 in.

[1] NIST, SI Units – Length.

FIELD NOTES / TEST FEEDBACK

Measure from one datum

Choose a repeatable reference: a marked centerline, an established face or an approved elevation benchmark. Label it A and keep dimensions tied to it.

Record what the tape actually touched. If you measure to the outside of a pipe but need its centerline, document the required geometric correction using the measured or specified outside diameter.

Use an independent overall measurement to check a chain of smaller measurements. Record offsets in separate horizontal and vertical directions before combining them.

Include obstruction envelopes, insulation space and maintenance access on the field sketch when required by the design. A centerline route alone does not describe all the space the assembly occupies.

Datum-to-tee: 24 in. Tee-to-end: 72 in.
Expected overall: 96 in.
If the independent overall is 97 in, reconcile the one-inch difference before cutting.

FIELD CHECK Write the date, drawing revision, instrument and any access limitation beside the survey.

FIELD NOTES / TEST FEEDBACK

Identify the actual component

A nominal pipe-size label is an identifier, not a promise that the measured outside diameter equals the label. Record material, size designation, wall designation and connection type.

For tubing, record the specified outside diameter and wall thickness. Do not substitute a pipe product merely because its description contains a similar number.

Use the exact manufacturer dimensional drawing for the component being installed. Charlotte Pipe, for example, publishes dimensional catalogs for its own pressure pipe and fitting families. [2]

Keep a component schedule: item tag, manufacturer, part number, material, size, end connections, required rating and dimensional source. Leave unknown values visibly unresolved instead of filling them from memory.

FIELD CHECK No pressure/temperature ratings or universal fitting dimensions are reproduced in this guide.

[2] Charlotte Pipe, Pressure Dimensional Catalog.

FIELD NOTES / TEST FEEDBACK

GEOMETRY

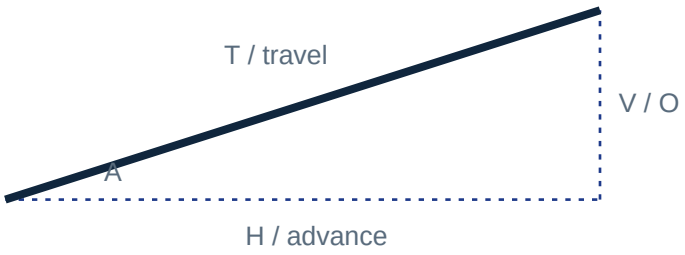
Solve the triangle first

Use a right triangle when two measured components are perpendicular. Name the perpendicular legs H and V, and the diagonal T.

T = sqrt(H² + V²). If T and H are known, V = sqrt(T² – H²). A negative quantity inside the square root means the inputs or the assumed geometry do not agree.

For an angle measured from H, sin(angle) = V/T and tan(angle) = V/H. Keep the calculator in degrees when using degree-marked pipe angles. [3]

The triangle describes geometric separation. It does not include elbow center-to-end deductions or joint allowances.



H = 12 in and V = 9 in.
T = sqrt(144 + 81) = 15 in.
Check: the diagonal is longer than either leg.

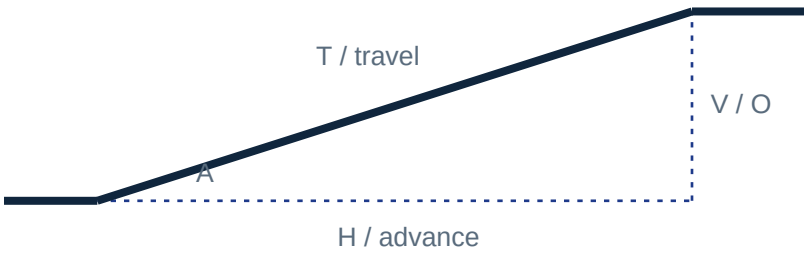
[3] Right-triangle identities; independent example.

Two-bend offset geometry

For parallel entering and leaving centerlines, define O as the perpendicular offset. Let angle A be the direction change of each bend. T is the diagonal distance between the theoretical bend-center intersections, and R is the advance along the original run direction.

From the right triangle: T = O / sin(A), and R = O / tan(A). These formulas apply to ideal centerline geometry with equal opposing bend angles.

Choose the bend angle from the approved layout and available components. A small angle needs a longer travel and more run space. Actual fitting bodies and joint requirements must fit within that geometry.



O = 10 in, A = 30°.
T = 10 / 0.5 = 20 in.
R = 10 / tan(30°) = 17.3205 in.
The 20 in travel is not the pipe cut length.

FIELD CHECK Before using the formula, confirm the two entering/leaving runs really are parallel.

Offset factors you can check

Multiply perpendicular offset O by the appropriate factor. These values were calculated directly from 1/sin(A) and 1/tan(A); they are not copied fitting dimensions.

The factors are rounded for quick checking. For a final layout, calculate from the angle at full precision, then round the final reported dimension.

Both answers have the same length unit as O. If O is in millimeters, travel and advance are in millimeters.

Bend angle	Travel × O	Advance × O
15°	3.8637	3.7321
22.5°	2.6131	2.4142
30°	2.0000	1.7321
45°	1.4142	1.0000
60°	1.1547	0.5774

For O = 8 in at 22.5°:
Travel ≈ 8 × 2.6131 = 20.9048 in.
Advance ≈ 8 × 2.4142 = 19.3136 in.

FIELD CHECK A 45° offset has advance equal to offset; its diagonal travel is about 1.4142 times the offset.

Worked 45-degree offset

Suppose the approved centerline offset is 18 in and the two direction changes are 45°. Establish the theoretical bend-center intersections on the sketch first.

Calculate travel and advance before selecting a straight pipe cut. Then retrieve the actual takeoff along the travel leg for each fitting.

Joint gaps in this example are invented for arithmetic practice, not a recommendation for any welding process. Real gaps come from the applicable procedure.

Travel = 18 / sin(45°) = 25.4558 in.
Advance = 18 in.

Assume verified travel-leg takeoffs of 4 in each, and two specified gaps of 1/8 in:
Cut = 25.4558 – 4 – 4 – 0.125 – 0.125
Cut = 17.2058 in, or 17-3/16 in to nearest 1/16.

FIELD CHECK Reconstruct the unrounded chain: cut + takeoff A + takeoff B + both gaps must equal travel.

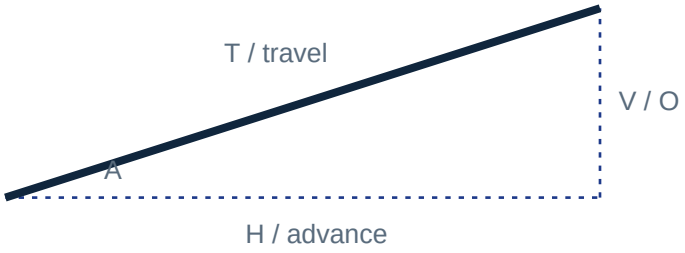
FIELD NOTES / TEST FEEDBACK

A rolling offset uses two triangles

A rolling offset moves sideways and changes elevation. First combine the two perpendicular offsets: true offset O = sqrt(side² + rise²). Then use O in the ordinary two-bend offset formula.

Define the roll angle from the side/horizontal component toward the vertical component: roll = atan2(rise, side). State your viewing direction and sign convention so the assembly cannot be mirrored by accident.

The roll angle orients the offset plane. It is not the elbow bend angle. Those two angles answer different questions.



Side = 8 in, rise = 6 in.
True offset = sqrt(64 + 36) = 10 in.
Roll from the side axis = atan2(6,8) = 36.8699°.
Using 45° bends: travel = 10 / sin(45°) = 14.1421 in; advance = 10 in.

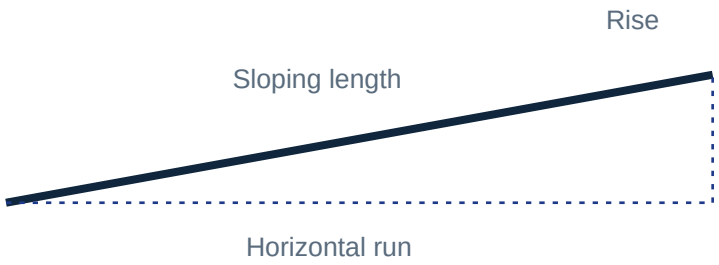
FIELD CHECK Check side and rise independently on the finished layout; the correct diagonal alone can still be in the wrong plane.

Slope is rise over horizontal run

Slope = vertical change / horizontal run. Multiply that ratio by 100 to express percent slope. Keep both values in the same unit.

An instruction stated as inches per foot describes vertical change for each foot of horizontal run. It is not degrees. The sloping pipe length is slightly longer than the horizontal run.

Use only the slope specified for the system. The example below teaches arithmetic and does not prescribe a drainage or process-piping slope.



Specified example: 1/4 in per ft for 16 ft.
Vertical change = 16 × 1/4 = 4 in.
Horizontal run = 192 in.
Percent = 4 / 192 × 100 = 2.0833%.
Sloping length = sqrt($192^2 + 4^2$) = 192.0417 in.

FIELD CHECK Mark the high and low ends. Do not infer flow direction from the way a drawing happens to face.

Build a dimensional chain

Start with a span whose reference points are explicit. Break it into every physical contribution between those points: fitting takeoffs, the pipe itself, required gaps, and any other specified pieces.

For the simple butt-end example here: C-C span = takeoff A + gap A + pipe cut + gap B + takeoff B. Rearranging gives the pipe cut.

Socket engagement, threads, flange faces and grooved connections use different physical reference points. Draw their chain instead of applying a universal deduction rule.

Dimensions in a catalog may use different datums. Read the sketch and notes attached to the dimension, not just its letter.



C-C = 48 in. Takeoffs = 4 in and 5 in.
Specified example gaps = 1/8 in at each end.
Cut = 48 – 4 – 5 – 1/8 – 1/8 = 38-3/4 in.

FIELD CHECK Kerf is a stock-use allowance; it is not automatically a joint gap in the finished assembly.

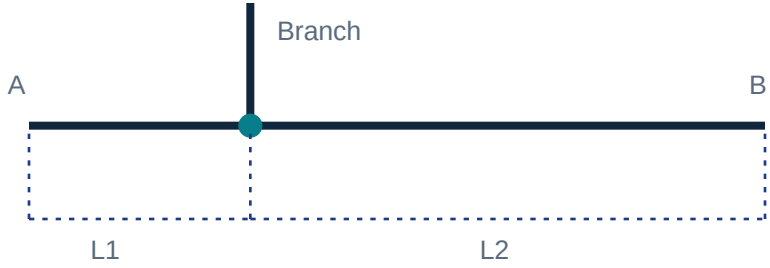
A tee creates separate sections

Give the two run sections and the branch separate identities. A tee center is a junction shared by those sections, not a floating symbol.

If the straight run contains lengths L1 and L2 measured from its end references to the tee center, the tee fraction along that run is L1 / (L1 + L2). With several tees, use cumulative section lengths divided by the run total.

That ratio controls a proportional drawing. Pipe cut lengths still need fitting takeoffs and the correct end references.

When a tee moves on a sketch, inspect every connected branch, flange and nearby dimension. A visually connected line is not proof that all dimensions have been updated.



Run total = 96 in; first section = 24 in.
Tee fraction = 24 / 96 = 0.25.
The tee belongs one quarter of the way along the run.
For 24, 30 and 42 in sections, tees lie at 25% and 56.25%.

Flanges: face, center and symbol

A flange face is a physical reference surface. A flange symbol is only a drawing convention. Neither its line thickness nor its apparent width on screen provides a fabrication dimension.

In this app, an end flange is drawn as one line across the pipe. The normal flange option is drawn with two lines. Confirm the drawing legend before treating those marks as a bill of materials.

Record the actual flange type, size, material, class, facing and mating details from the specified component documents. Include required face-to-face space and gasket contributions in the correct dimensional chain.

Bolt torque, gasket selection and assembly sequence are joint-specific. Use the approved joint assembly procedure rather than a generic number.

FIELD CHECK If the field measurement reaches the sealing face, label it F. Do not silently move its reference to the flange back or pipe end.

FIELD NOTES / TEST FEEDBACK

Valve layout includes access

Record the valve tag and exact ordered model before using a face-to-face dimension. Different bodies, end connections and actuators can occupy different envelopes.

Show intended orientation and any specified flow arrow. Include operator travel, actuator clearance and maintenance removal space in the layout review.

Draw a dimension chain through the valve assembly using the actual ends or faces. Do not replace the valve with a guessed straight-pipe length in the cutting list.

Keep the functional specification separate from the geometry: a valve that fits between two faces is not necessarily suitable for the fluid, temperature or required service.

> If an overall face-to-face envelope is 40 in and a verified valve assembly occupies 12 in of that chain, 28 in remains for the other components and specified joint contributions. That remainder is not automatically one pipe cut.

FIELD CHECK Confirm the valve can be operated and maintained after adjacent pipe is installed.

FIELD NOTES / TEST FEEDBACK

Tube bends: geometry and tooling

For an ideal circular centerline bend, use centerline radius R and bend angle A in degrees. Arc length B = pi × R × A / 180. The setback from a theoretical corner to a tangent point is S = R × tan(A/2).

For two straight legs meeting at that corner, the ideal developed-length gain is G = 2S – B. This is geometric bookkeeping, not a universal bender deduct or arrow setting.

Actual tooling marks and springback corrections depend on the bender, material and tube. Use the manufacturer instructions for the exact tool. [4]

R = 3 in, A = 90°.
S = 3 × tan(45°) = 3 in.
B = pi × 3 × 90 / 180 = 4.7124 in.
G = 6 – 4.7124 = 1.2876 in.

FIELD CHECK Keep CLR, outside radius and inside radius distinct. Do not use a pipe-fitting takeoff as a tubing bend radius.

[4] Swagelok tool guidance; equations and example independently developed.

FIELD NOTES / TEST FEEDBACK

Clock the next bend

A bend angle changes direction within a plane. Clocking rotates that plane around the tube axis before the next bend. Record both values.

Choose a viewing direction, such as looking from the datum end toward the free end. Keep it fixed on the sketch and in the work sequence.

Draw a reference stripe and number the bends in order. Note whether each measurement reaches a tangent, a theoretical center intersection or a finished end.

Before forming the part, check that the chosen sequence can be held in the actual tool without interference. Manufacturer mark conventions take priority over a generic sketch.

Bend record example: B1 = 45°, plane 0°. B2 = 45°, plane rotated 30° clockwise as viewed from datum A. The 30° plane rotation does not replace either 45° bend angle.

FIELD CHECK A mirrored part may have every individual length correct. Verify orientation with the marked datum still visible.

FIELD NOTES / TEST FEEDBACK

Read an ISO as a map

An isometric drawing communicates connections and direction. Unless the drawing explicitly states otherwise, do not measure its printed line lengths with a ruler to obtain fabrication dimensions.

Trace from a known datum or equipment connection. Number each straight section and identify every change of direction, junction, termination and component.

At a crossing, check whether the lines actually join. A tee or connection mark is different from one line passing another.

Keep the north arrow or orientation reference separate from the page edge. Confirm elevations, slope direction and branch orientation with the associated views.

FIELD CHECK Before making a cut list, explain the complete route aloud from one end to the other. Every unexplained jump is a drawing question to resolve.

FIELD NOTES / TEST FEEDBACK

APP PRACTICE

Build the route in the app

In ISO Drawing with PIPE LINE active, tap a start point, release, then tap the next point. Continue with separate taps to add sections.

To branch, tap the body of an existing pipe. The app inserts a tee and splits the run into two sections. The next grid tap starts the branch from that junction.

Tap an existing endpoint to make it the active drawing point. A one-finger swipe pans the view; two-finger pinch changes zoom. The view also follows newly added pipe near the viewport edge.

Drag an endpoint toward an edge to extend the pipe while the view pans. Release to stop. Select END FLANGE for the single-line terminal symbol.

> Try a simple run, one tee, one branch and one end flange. Select each section in the list and check that the corresponding pipe highlights.

FIELD CHECK This edition describes the v87 drawing controls; report any phone-specific behavior that differs.

FIELD NOTES / TEST FEEDBACK

APP PRACTICE

Make dimensions easy to trace

Select the intended pipe before editing its measurement. Check whether the value is C-C, E-E, C-F, F-F or C-E, and keep that basis on your field notes.

Drag the dimension box to clear space. Its dark blue dotted lines should follow the box while their pipe ends stay on the section. There is no extra middle connector.

Use DIM – or DIM + for size, FLIP DIM for the opposite side, and ROTATE DIM to change text orientation. Keep a little room between neighboring boxes.

Changing measured straight-run section lengths repositions tees proportionally. Unmeasured sections use inferred drawing lengths until you enter real measurements; replace those estimates before fabrication.

FIELD CHECK Trace both dotted endpoints for every label. A readable number attached to the wrong section is still an error.

FIELD NOTES / TEST FEEDBACK

Prepare a spool record

A useful spool record lets another person rebuild your reasoning. Include the spool identifier, line/service identifier, drawing revision, material and component schedule.

For each section, record its reference-to-reference measurement, both fitting takeoffs, any specified joint contributions, the resulting cut length and the rounding increment.

Keep the raw dimension and final cut in separate columns. Include the source for each non-geometric value, such as a manufacturer part drawing or approved procedure.

Identify match marks and the viewing direction for clocked branches. Record which dimensions are verified, inferred or still awaiting confirmation.

Section	Raw span	Cut	Source/status
S1	C-C 48 in	38-3/4 in	Example only
S2	C-F ___	___	Await face datum
S3	C-C ___	___	Verify branch plane

FIELD CHECK Do not mark a spool ready solely because every cell contains a number.

FIELD NOTES / TEST FEEDBACK

Joining details are not guesses

Check the specified joint type before preparing a pipe end. Welding, threading, socket connections, grooving and mechanical couplings have different dimensional and assembly requirements.

Keep component identification and traceability intact through layout and cutting as required by the project. Record substitutions through the established approval process.

Pressure-test values, hold times, temperature limits, torque values and welding parameters do not appear in this guide. Retrieve them from the controlling system documents and approved procedures.

An app calculation verifies arithmetic only. It cannot determine material compatibility, code acceptance or whether a field assembly is safe to energize.

FIELD CHECK Pause the handoff when component markings, drawing callouts and the material record disagree. Resolve the discrepancy with the responsible person.

FIELD NOTES / TEST FEEDBACK

Plan the work before opening a line

Identify the actual service and hazards before work begins. Stored pressure, thermal energy and unexpected re-energization need the applicable site energy-control process. OSHA describes hazardous-energy controls for covered work; the governing requirements depend on the work and industry. [5]

Hot work also requires control of ignition and exposure hazards. OSHA identifies fire, fumes and radiation among the hazards associated with welding and cutting. Use the applicable permit, supervision and protective measures. [6]

This page is a planning prompt, not an isolation or hot-work procedure. Obtain the site-authorized instructions and trained personnel for the task.

FIELD CHECK Confirm who authorizes the work, which system is involved, how its condition is verified, and which current procedure applies.

[5] OSHA hazardous energy overview. [6] OSHA welding hazards overview.

FIELD NOTES / TEST FEEDBACK

QUALITY CHECKS

Check the assembly before release

Use independent measurements for the overall dimensions and branch locations. Confirm that the sum of the documented section chains agrees with the overall span.

Review orientation, flange faces, valve access, connection types and any required slope against the approved layout.

Make labels legible on the drawing and on the record. Note revision changes clearly so an older cut list cannot be mistaken for the current one.

Leave unresolved items open. Completion of a drawing or an app checkbox is not acceptance of fabrication, inspection or testing. Record those approvals separately according to the project.

Review item	Record
Overall dimensions	Checked by / date
Tee and branch positions	Datum / result
Faces and orientations	Drawing revision
Component identity	Schedule reference
Inspection or test status	Approved record ID

FIELD CHECK Save a copy of the reviewed drawing with its revision and supporting measurements.

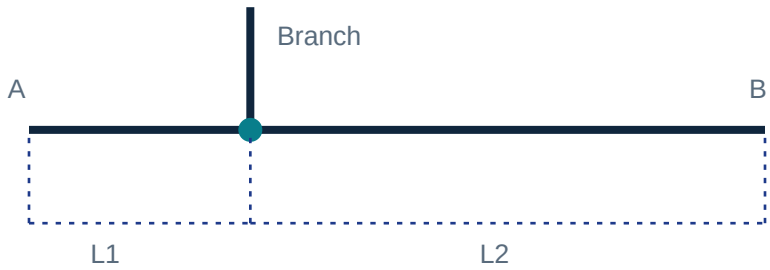
PUT IT TOGETHER

Worked spool: two run sections

This practice spool has two straight run spans meeting at a tee. All takeoffs and joint gaps below are hypothetical values supplied for the exercise.

Datum A to tee center = 30 in; tee center to datum B = 66 in. The total is 96 in, so the tee center should appear 31.25% of the way from A to B.

S1 uses takeoffs of 4 in and 3 in with two 1/8 in gaps. S2 uses takeoffs of 3 in and 5 in with two 1/8 in gaps.



S1 cut = 30 – 4 – 3 – 0.25 = 22.75 in.
S2 cut = 66 – 3 – 5 – 0.25 = 57.75 in.

Check S1: 4 + 0.125 + 22.75 + 0.125 + 3 = 30.
Check S2: 3 + 0.125 + 57.75 + 0.125 + 5 = 66.

Enter 30 in and 66 in as the run-section C-C values. Do not enter the cut lengths as C-C.

FIELD CHECK The branch needs its own span and takeoff calculation; neither run-section result supplies its cut length.

Working vocabulary

Term	Meaning in this guide
Datum	A stated reference for measurement.
Centerline	The geometric axis of a pipe or tube.
Takeoff	Distance from a stated fitting reference to its connection end or face.
Travel	Diagonal span in the offset triangle.
Advance	Span along the original run direction.
True offset	Combined perpendicular displacement in the offset plane.
Roll	Orientation of an offset plane.
CLR	Radius to the centerline of a circular bend.
Tangent	Where a straight centerline meets a bend arc.
Kerf	Material removed by the cutting process.
Spool	An identified assembly of pipe and components.
Witness line	Line tying a dimension to a reference point.

FIELD CHECK Use the project legend when its terminology differs. State the definition beside an ambiguous measurement.

FIELD NOTES / TEST FEEDBACK

REFERENCE

Sources and edition notes

This guide uses original explanations, independently calculated examples and newly drawn diagrams. It does not reproduce Audel text, page layouts, illustrations or tables. Sources below support unit definitions, basic geometry or direct readers to controlling technical information.

Original chapters provide geometry and reference guidance. The practical supplement adds selected model-specific factual settings with attribution; it does not reproduce a complete manufacturer manual. Mention does not imply endorsement.

Edition 0.4 is for field reference and app review. Before commercial publication, have a qualified technical reviewer check the formulas, examples, terminology and scope. Confirm the final content and illustration rights, and record the reviewer and revision date.

Sources checked October 9, 2026 (UTC).

[1] NIST: SI Units – Length
https://www.nist.gov/pml/owm/si-units-length
[2] Charlotte Pipe: Pressure Dimensional Catalog
https://www.charlottepipe.com/technical-hub/pressure-dimensional-catalog
[3] OpenStax: Right Triangle Trigonometry
https://openstax.org/books/precalculus-2e/pages/5-4-right-triangle-trigonometry
[4] Swagelok: Tube Fitter’s Manual resource
https://www.swagelok.com/en/resources/tube-fitting-manual
[5] OSHA: Control of Hazardous Energy
https://www.osha.gov/control-hazardous-energy
[6] OSHA: Welding, Cutting and Brazing Hazards
https://www.osha.gov/welding-cutting-brazing/hazards-solutions

FORMULA WORKSHOP

Work a formula, one step at a time

1. Sketch the part. Write the quantity you need and its units. Name every known value on the sketch; do not start by entering numbers into a calculator.

2. Select an equation whose assumptions match that sketch. Rearrange it for the unknown before substituting values. Use parentheses around a whole numerator or denominator.

3. Convert to one unit system. Set trigonometry to degrees for degree angles. Evaluate brackets, powers and roots before multiplication and division. Keep extra digits until the final answer.

4. Put the answer back into the original equation and estimate whether its size makes sense. Show the equation, substitution, result and check in the job record.

Find an offset when travel T = 20 in and angle A = 30°.
T = O / sin(A), so O = T × sin(A).
O = 20 × sin(30°) = 20 × 0.5 = 10 in.
Back-check: 10 / sin(30°) = 20 in.

Find A when O = 10 in and T = 20 in:
A = arcsin(O/T) = arcsin(0.5) = 30°.
Use inverse sine, not 1/sin.

FIELD CHECK An arcsin input outside -1 to +1 signals an invalid ratio for this triangle.

Circumference and equal stations

Use actual outside diameter D for a pattern wrapped around the outside surface. Circumference C = pi × D. Radius r = D/2. A nominal pipe designation is not an outside diameter.

For N equal stations, spacing s = C/N and angular step = 360°/N. Measure distances along the paper baseline, not chord distances across the circular end.

Station k sits at k × C/N. Calculate each station from zero rather than repeatedly adding a rounded spacing. The last station closes on the first.

Illustrative actual OD = 4.000 in; N = 16.
C = pi × 4 = 12.56637 in.
s = 12.56637 / 16 = 0.785398 in.
Angular step = 22.5°.
Station 4 = C/4 = 3.14159 in, at 90°.
Station 8 = C/2 = 6.28319 in, at 180°.
Station 16 = C = 12.56637 in, at 360°.

FIELD CHECK If you measure the circumference directly with paper, use that measured working length for folding. Keep any tape tab outside it.

FIELD NOTES / TEST FEEDBACK

Estimate pipe weight

Prefer documented assembly weights or a verified weighing method. A geometric estimate needs actual dimensions, material weight density and every attached component.

For straight hollow round pipe: inside diameter d = D – 2t; metal area A = pi × (D^2 – d^2)/4; volume V = A × L; weight W = V × gamma. D, d, t and L must use the same length unit.

Here gamma is weight per unit volume, not mass density. If mass is given in kilograms, convert mass to force using W = m × g; at g = 9.80665 m/s^2, 1000 kg corresponds to 9.80665 kN.

Add fittings, flanges, valves, attachments, contents and other carried items. Include suspended rigging in the hoist load; identify which weights pass through each sling. Unknown contents or weights must be resolved before lifting.

Practice inputs only: D = 4 in, t = 0.25 in,
L = 120 in, gamma = 0.283 lbf/in^3.
d = 3.5 in. A = pi(16 – 12.25)/4 = 2.94524 in^2.
V = 353.429 in^3. W = 100.020 lbf.
Add assumed attachments of 40 lbf: 140.020 lbf.
This estimate is not a verified lift weight.

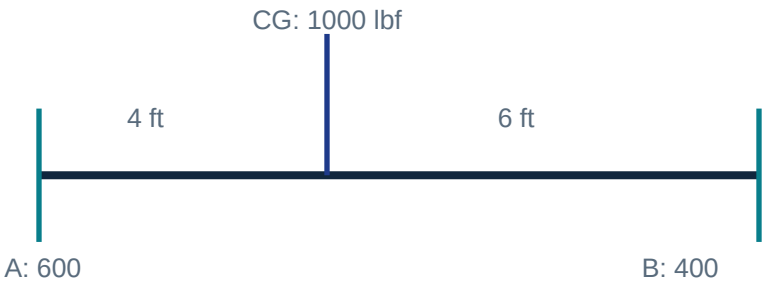
FIELD CHECK A hollow pipe calculation is wrong for a pipe full of liquid, plugged with solids or carrying uncounted fittings.

Find the center of gravity

For known component weights along one axis, xCG = sum(Wi × xi) / sum(Wi). Choose a common datum and locate each component at its own center of gravity.

For two vertical reactions at A and B, spacing S, with CG a distance x from A: VB = W × x/S; VA = W – VB. The support nearer the CG takes more vertical load.

This is a static two-support model. The CG must lie between the supports. A full assembly also needs transverse and vertical CG, stability, attachment strength and clearance evaluated. The hook must be positioned over the combined suspended CG. [8]



600 lbf centered at x = 2 ft; 400 lbf at x = 7 ft.
Total W = 1000 lbf.
xCG = (600 × 2 + 400 × 7)/1000 = 4 ft.
If A is at 0 ft and B at 10 ft:
VB = 1000 × 4/10 = 400 lbf.
VA = 1000 – 400 = 600 lbf.
Check moment about A: 400 × 10 = 1000 × 4.

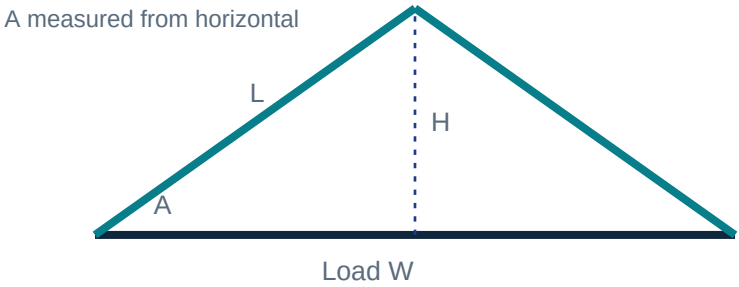
[8] Static equilibrium principles; original example.

Two equal sling legs

For a centered, level load W on two equally loaded straight sling legs, measure angle A from the horizontal load line. Each leg carries vertical share W/2. Its tension T = W / (2 × sin(A)).

If the angle is measured from vertical instead, use cosine. Do not mix the two angle conventions. The included angle at the hook is a third, different angle.

When measuring sling length L from load attachment to hook and vertical rise H, sin(A) = H/L, so T = (W/2) × L/H. These are static forces; shock and acceleration are excluded. [7]



W = 2000 lbf; A = 60° from horizontal.
T = 2000/(2 × 0.866025) = 1154.70 lbf per leg.
At 45°: T = 1414.21 lbf per leg.
At 30°: T = 2000 lbf per leg.
Check: 2 × 1154.70 × sin(60°) = 2000 lbf.

FIELD CHECK The angles shown are arithmetic examples, not approved configurations. Lift-All advises against angles below 30° from horizontal.

[7] Lift-All angle guidance; examples calculated independently.

Unequal sling loading

Do not divide weight equally when the CG is off center. First calculate each vertical share, then Ti = Vi/sin(Ai) for the angle of that leg from horizontal.

For a level rigid load with same-elevation attachment points and hook above the CG, let a and b be the horizontal distances from A and B to the hook vertical, and H the rise. Then LA = sqrt($a^2 + H^2$), LB = sqrt($b^2 + H^2$).

VA = W × b/(a+b); VB = W × a/(a+b). TA = VA × LA/H; TB = VB × LB/H. The inward horizontal forces must balance in this ideal model. Different elevations or out-of-plane geometry need a different analysis.

W = 3000 lbf; a = 4 ft; b = 6 ft; H = 6 ft.
VA = 1800 lbf; VB = 1200 lbf.
LA = sqrt(52) = 7.2111 ft; LB = sqrt(72) = 8.4853 ft.
TA = 1800 × 7.2111/6 = 2163.33 lbf.
TB = 1200 × 8.4853/6 = 1697.06 lbf.
Horizontal check: 1800 × 4/6 = 1200 × 6/6.
Both equal 1200 lbf inward.

FIELD CHECK Neither the average tension nor the smaller tension can be used to select both legs.

FIELD NOTES / TEST FEEDBACK

Turn forces into a lift review

A tension calculation is a demand, not a sling rating. Compare demand with the manufacturer rating for the actual sling, hitch, angle and service conditions. Read whether a tag rates one leg or the complete assembly.

A rated two-leg assembly chart may already include the angle effect. Do not apply the angle factor twice. Do not assume four legs share load equally or that a basket always doubles usable capacity.

Inspect identification and condition before use. Remove defective gear from service; protect slings from damaging edges. Check shackles, hooks, attachment points and the supporting structure as well as the sling. [9]

A qualified lift planner must address load control, CG uncertainty, dynamic effects, hitch stability and any manufacturer reductions. Do not invent a general safety multiplier or use breaking strength as working load limit.

Calculated leg demand = 2163.33 lbf.
An assumed 2000 lbf allowable leg capacity is insufficient.
An assumed 3000 lbf capacity passes only this numerical comparison; the actual configuration, hardware and lift plan still require verification.

FIELD CHECK Keep people clear of suspended loads. Stop if weight, CG, attachment rating or sling identification is unknown.

[9] OSHA 1926.251 and manufacturer rating instructions; this is a training calculation, not a lift plan.

Oxy-acetylene CUTTING setup

These pages are for OXY-ACETYLENE CUTTING of suitable carbon steel, not fusion welding, heating or brazing. Cutting tips have a central cutting-oxygen jet plus preheat ports. Welding tips and fuel-gas substitutions use different data.

Identify the exact cutting tip series and size marked on the tip. Verify compatible torch and attachment, oxygen and acetylene regulators, hose sizes, tip condition, flashback protection and required gas flow. Match that marked series to the correct chart; never average or mix charts.

Secure cylinders upright; protect oxygen gear from oil and grease. Inspect hoses and connections, leak check and purge each gas separately according to the equipment manual. Observe hot-work controls, fire watch, ventilation and suitable PPE. [10,11]

Delivery-pressure settings are read on the LOW-PRESSURE (working) gauges under the manufacturer-specified flow conditions. The cylinder contents gauges indicate supply pressure, NOT the setting to use. Never use acetylene above 15 psig. [10]

FIELD CHECK Before adjusting any regulator, confirm: fuel is ACETYLENE, tip type and size match the chart, and torch/cutting attachment is compatible.

[10] OSHA 1910.253. [11] Manufacturer oxygen-fuel setup guidance. Complete equipment manual governs use.

FIELD NOTES / TEST FEEDBACK

CUTTING chart: Harris V1-101

Harris V1-101 single-piece acetylene HAND-CUTTING tips for relatively clean steel, compatible with identified V-Series and Victor-style torches. Manufacturer chart PN 9700378 (08/2023), page 6. [15]

The oxygen pressure is for the cutting operation; oxygen and acetylene delivery values are in PSIG for CUTTING. They are NOT welding-tip settings. Values apply only to the exact tip series and compatible equipment. Recheck the current Harris data for torch, hose and required flow.

Steel thickness in	Tip size	Oxygen psig	Acetylene psig
To 1/8	000	20-30	5
1/8 to 5/16	00	20-30	5
5/16 to 5/8	0	30-40	7
5/8 to 1	1	40-50	7
1 to 2	2	45-55	8
2 to 3	3	50-60	9
3 to 4	4	60-75	10

FIELD CHECK For thicker cuts the source also lists size 5 (4-6 in: O2 70-80, acetylene 12) and size 6 (6 in: O2 75-80, acetylene 15); special hose requirements apply. Never exceed approved working pressure.

[15] The Harris Products Group, V1-101 general preheat acetylene cutting tips, PN 9700378, 08/2023, page 6. Selected facts re-tabulated, not a complete copy.

CUTTING chart: Victor Type 101

Victor Type 101 acetylene hand/machine CUTTING tip operating data from ESAB. This is a DIFFERENT series from Harris V1-101 and Smith SC12; do not interchange regulator settings. Values below are selected ESAB listings. [16]

Oxygen column means CUTTING-oxygen pressure (psig); acetylene column means fuel-gas pressure (psig). ESAB also publishes separate PREHEAT-oxygen values, so check the full chart and the specific torch configuration.

Steel thickness in	Tip size	Cut O2 psig	Acetylene psig
1/4	00	20-25	3-5
3/8	0	25-30	3-5
1/2	0	25-30	3-5
3/4	1	30-35	3-5
1	2	40-45	3-7
1-1/2	2	40-45	3-7
2	3	40-45	4-9

FIELD CHECK Separate PREHEAT-oxygen pressure values in Victor charts are for compatible three-hose torch setups. With a two-hose cutting torch, preheat oxygen shares the cutting-oxygen regulator supply. Do not try to dial in a separate preheat value on a two-hose torch.

[16] ESAB/Victor, Type 101 acetylene cutting tips, Operating Specifications. Selected rows reflect listed tip/part variants; consult the exact marked tip and the full source.

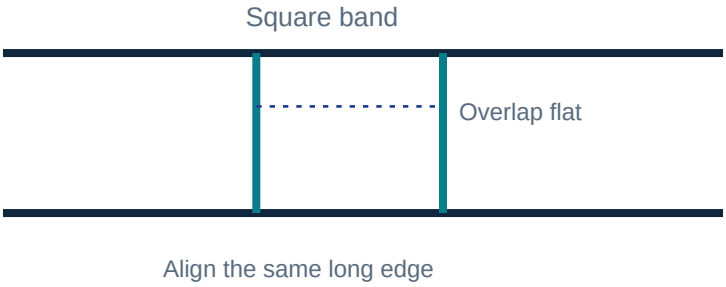
Use a pipe wraparound

1. Secure the cold pipe against rolling. Clean the layout band. Select an undamaged wrap long enough to overlap and wide enough to stay square; obey its temperature limit. A wraparound is a marking guide, not a torch heat shield. [14]

2. Mark the desired cut location from a verified datum. Bring one straight long edge of the wrap to that mark and wrap snugly around the pipe without twisting.

3. At the overlap, align the long edge with itself. Hold the layers flat. The edge must meet continuously with no step; a spiral or skewed overlap makes a sloping cut line.

4. Trace along that same edge around the full circumference, repositioning your hand without shifting the wrap. Check closure and measure from the datum at several positions. Remove the wrap before hot work.



FIELD CHECK A dented or out-of-round pipe can prevent a clean square wrap. Resolve the geometry rather than forcing a distorted guide.

[14] Jackson Safety tool purpose; original marking workflow.

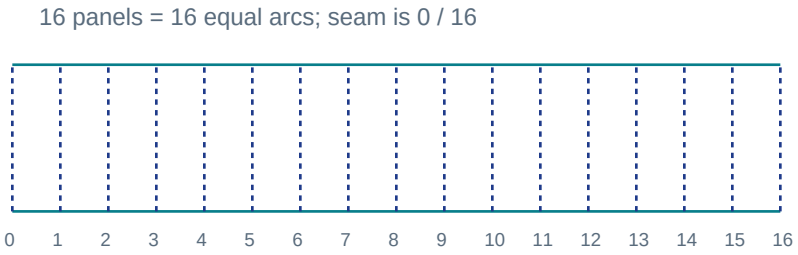
Cut and fold register paper

Use dry, unstretched register paper wide enough for the largest ordinate plus a holding margin. Keep it away from heat. If it is too narrow, use wider drafting paper; do not omit part of the curve.

1. Wrap the strip once around the actual branch OD with square overlapping edges. Mark where the starting end meets the strip. Unwrap and cut at that mark: the working length is exactly one circumference. Add any fastening tab later.

2. Bring the two short ends together and crease: two equal panels. Keep folding the stacked strip end-to-end three more times, aligning edges each time. Four halvings give 16 panels. Unfold completely.

3. Number the edges and creases 0 through 16. Draw each crease straight across the width, perpendicular to a common long-edge baseline. These are the ordinate station lines. Stations 0 and 16 become the same seam when wrapped.



Fold count: 1 → 2 panels; 2 → 4; 3 → 8; 4 → 16.
Each panel represents 360°/16 = 22.5°.
To use 12 stations, measure C/12 instead; repeated halving does not produce 12 equal divisions.

FIELD CHECK Folding locates stations only. It does not determine the saddle heights. Calculate and mark those after unfolding.

Calculate a 90-degree saddle

Scope: round branch meeting a round main at 90°, with intersecting axes, branch radius r no greater than main radius R. Use actual outside radii. This is the outside-surface branch-end profile, not the opening in the main.

Look down the branch axis. Station 0 lies in the direction of the main axis; station 4 is 90° around the branch. At angle A, the side displacement is $y = r \times \sin(A)$. The main surface height is $z = \sqrt{R^2 - y^2}$.

The lowest contact height is $z_{min} = \sqrt{R^2 - r^2}$. Measure each cutback ordinate O from the plane of the longest fingers, toward the retained pipe: $O = \sqrt{R^2 - (r \times \sin(A))^2} - \sqrt{R^2 - r^2}$.

On paper, use the long edge as that longest-tip baseline. Plot O on each station into the retained side and connect with a fair curve. The region between baseline and curve is waste. Do not cut the stacked folded paper.

FIELD CHECK This formula excludes offsets, oblique branches, bevels, root gap and reinforcement. Pressure-piping branch design and weld preparation require the approved detail.

FIELD NOTES / TEST FEEDBACK

Worked saddle: 4 OD onto 8 OD

Illustrative actual ODs are 4.000 and 8.000 in, not nominal pipe sizes. r = 2 in; R = 4 in; C = 12.56637 in; station spacing = 0.785398 in.

At station 2, A = 45°: $y^2 = (2 \times \sin 45°)^2 = 2$. O = sqrt(16 – 2) – sqrt(16 – 4) = 0.277556 in.



Station(s)	Angle example	O, inches
0, 8, 16	0° / 180° / 360°	0.535898
1, 7, 9, 15	22.5°	0.461992
2, 6, 10, 14	45°	0.277556
3, 5, 11, 13	67.5°	0.083543
4, 12	90° / 270°	0.000000

FIELD CHECK All ordinates are nonnegative. The two longest fingers are at stations 4 and 12. The two deepest cutbacks are at 0 and 8. Diagram is schematic, not a full-size template.

Equal-size saddle and more stations

When R = r, sqrt(R² – r²) = 0, so O = R × abs(cos(A)). The absolute value keeps the cutback nonnegative around the full circle.

For actual equal ODs of 4 in, R = 2 in. At 0°, 22.5°, 45°, 67.5° and 90°, the cutbacks are 2.0000, 1.8478, 1.4142, 0.7654 and 0.0000 in. Repeat by symmetry around the other quadrants.

Large pipes or rapidly changing curves may need more stations. Another halving produces 32 panels at 11.25° each. Recalculate ordinates at those angles; do not just average neighboring heights.

Equal-size 90° saddles have sharp changes at the longest fingers. Preserve those intersections in the geometric pattern; do not round them away by eye. Fit and finish still follow the approved joint detail.

At A = 67.5° and R = 2 in:
O = 2 × abs(cos67.5°) = 0.765367 in.
For 32 stations, station 6 is 6 × 11.25° = 67.5°,
so it must reproduce that same ordinate.

FIELD CHECK The blank paper width must exceed the greatest ordinate plus a holding margin. A 2 in-wide strip cannot hold a 2 in ordinate and a useful margin.

FIELD NOTES / TEST FEEDBACK

Transfer, trim and check the saddle

1. With paper flat, connect calculated marks smoothly, preserving any genuine corner. Label main OD, branch OD, angle, station 0, retained side and baseline. Cut only the single-layer developed curve.

2. Wrap the paper around the branch, align stations 0 and 16 without adding overlap to the working circumference, and secure with a separate tape tab. Check that the baseline is square and the seam closes.

3. Align the station-0 direction with the main centerline plane. Trace the curve. Mark the scrap side clearly. Remove paper and wraparound before cutting.

4. Cut on the scrap side, leaving any finishing allowance required by the job. Trial-fit to a verified main or matching gauge and inspect the whole circumference. Apply the approved bevel and root gap separately.

5. Check axial length from the correct datum: this template references the longest fingers. A centerline-to-centerline dimension is not automatically the same length.

FIELD CHECK Practice on paper or scrap first. A close geometric fit does not qualify a pressure connection, establish reinforcement or approve a welding procedure.

FIELD NOTES / TEST FEEDBACK

REFERENCE

Practical supplement: sources

Practical sources identify the manufacturer cutting-tip series, rigging examples and measuring references. Cutting settings on pages 41-42 are distinct CUTTING-tip data, not welding instructions.

Saddle ordinates are derived from the ideal intersection of perpendicular circular cylinders. Verify a paper trial and the actual OD on the job; reinforcement and bevel detail are not supplied by these equations.

Illustrations and worked calculations in this guide are original. The linked manufacturer and standards publications govern ratings, pressure settings and approved procedures.

[7] Lift-All: Effect of sling angle
https://lift-all.com/product-files/catalog/effect-of-angle-chart.pdf
[8] OpenStax: Conditions for static equilibrium
https://openstax.org/books/university-physics-volume-1/pages/12-1-conditions-for-static-equilibrium
[9] OSHA 1926.251: Rigging equipment
https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.251

REFERENCE

Sources: oxy-fuel and layout

Use the current exact torch and cutting-tip documentation, rather than a setting from an unrelated tip or welding chart. The gas pressure seen on the delivery regulator may change under flow; follow the documented setup method.

Pages 41 and 42 use selected values for Harris V1-101 and Victor Type 101 CUTTING equipment only. For other brands, tip series, fuels and metal thickness, obtain the matching current manufacturer chart.

Smith SC12 is another acetylene cutting-tip family, not welding. It is not the chart used on these revised pages.

[10] OSHA 1910.253
https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.253
[11] Miller safe oxy-fuel setup
https://www.millerwelds.com/en-us/resources/knowledge-hub/welding-safety/10-steps-for-safe-oxy-fuel-torch-setup
[15] Harris V1-101 cutting tips (PN 9700378)
https://ch-delivery.lincolnelectric.com/api/public/content/bd29bc5b871a45aeb8ba5cf249542170?v=9a551e9d
[16] Victor Type 101 cutting tips, ESAB
https://esab.com/us/nam_en/products-solutions/product/gas-equipment/wear-parts/type-101-acetylene-cutting-tips/

Cutting: what the cut is doing

A cutting attachment adds a high-purity cutting-oxygen jet through its center opening. Preheat flames bring carbon steel to the ignition condition; oxygen oxidizes the metal and blows oxide from the kerf. A welding tip does not provide that same jet.

Square cuts depend on suitable tip size, clean orifices, constant stand-off, work surface condition, cutting-oxygen flow and a steady travel rate. A crooked kerf may be a tip, angle, oxygen-flow or travel-speed issue; diagnose from the manufacturer instructions.

For pipe, mark the cut with a verified wraparound. Check the pipe is cold, isolated, cleaned and secured. Account for the kerf and any field-finishing allowance in cut length; never assume the cut face is an approved weld bevel.

Follow the specific torch operating sequence for setup, light, cut, shutoff, separate bleeding and regulator release. Do not infer the sequence from this summary.

FIELD CHECK A suspected flashback, unstable flame, leak or damaged tip requires stopping work and using the documented response, not experimenting with pressure.

[10,11,15,16] Cutting and equipment-safety references.

FIELD NOTES / TEST FEEDBACK

Nominal pipe size versus OD

NPS is a designation, NOT the actual outside diameter for common sizes through NPS 12. Confirm pipe versus tubing before laying out a saddle, wraparound or support.

These are common nominal wrought-steel pipe outside diameters in inches. Verify actual product, applicable ASME B36.10/B36.19 data and material specification; do not substitute nominal size for a measured OD.

One nominal size may have several wall schedules. The outside diameter generally stays the same within a nominal size; the ID changes with wall thickness.

NPS	Actual OD in
1	1.315
1-1/2	1.900
2	2.375
3	3.500
4	4.500
6	6.625
8	8.625
10	10.750
12	12.750

FIELD CHECK For NPS 4 pipe the circumference is pi × 4.500 = 14.137 in, not pi × 4 = 12.566 in.

[17] ASME B36.10/B36.19 size designation. Selected standard steel pipe OD values.

Inside diameter and metal area

Given actual OD D and measured or specified wall thickness t, the approximate inside diameter is d = D - 2t, provided t is positive and less than D/2.

Flow opening area is Ai = pi × d²/4. Steel cross-section area is Am = pi × (D² - d²)/4. Units are square inches if diameters are inches.

The schedule label alone does not give an ID until the actual product dimensions and corrosion or machining allowances are known. This geometric area is not a pressure rating.

Illustrative OD 4.500 in, wall 0.237 in.
ID = 4.500 - 2 × 0.237 = 4.026 in.
Flow area = pi × 4.026² / 4 = 12.731 sq in (rounded).
Use the actual wall in the specification for a production calculation.

FIELD CHECK Do not confuse inside flow area with the steel metal area needed for weight estimates.

FIELD NOTES / TEST FEEDBACK

Parallel pipe and insulation clearance

For two bare round pipes of ODs D1 and D2, their minimum geometric centerline separation for just-touching outsides is (D1 + D2)/2. Add the required clear working space C to obtain centerline spacing S = (D1 + D2)/2 + C.

If the lines have insulation thicknesses I1 and I2, use S = (D1 + D2)/2 + I1 + I2 + C. Add any further clearance required for flanges, valve operation, weld access, movement and support details.

This is an envelope layout check, not a support design or thermal-movement analysis. Insulation can be removed locally for fittings; base spacing on the actual project arrangement.

6 in NPS pipe (OD 6.625) beside 4 in NPS pipe (OD 4.500).
Each insulated 1.5 in radial thickness, with 2 in free gap required.
S = (6.625+4.500)/2 + 1.5 + 1.5 + 2 = 10.5625 in center-to-center.

FIELD CHECK Confirm whether the specified clearance is between steel, insulation jacketing or projecting components.

FIELD NOTES / TEST FEEDBACK

Bolt circles and equal bolt spacing

Let a flange bolt circle have diameter B and N evenly spaced holes. Adjacent hole center spacing measured straight across is the chord c = B × sin(180°/N). The circumference arc spacing is pi × B/N.

A quick two-hole orientation has the top pair straddling the vertical centerline, if the drawing specifies that orientation. Actual clocking comes from the flange standard and project detail.

The bolt circle is a layout dimension; it does not determine bolt size, flange rating, gasket selection or allowable torque. Use a verified flange template for drilling or machining.

Illustrative bolt circle B = 7.500 in, N = 8.
Angle step = 360° / 8 = 45°.
Adjacent chord = 7.5 × sin(22.5°) = 2.870 in (approx).
Arc between holes = pi × 7.5 / 8 = 2.945 in.

FIELD CHECK Check from the actual drawing whether the flange is to be two-holed, one-holed or clocked to another orientation.

FIELD NOTES / TEST FEEDBACK

Flange assembly checks

Verify flange class, facing, gasket material, bolt grade and pipe specification from the approved isometric and line class. Align bolt holes and centerlines before bringing faces together.

Check gasket location, face condition, parallelism and the specified gap or insertion condition. Do not use bolt tightening to pull an ill-fitting pipe or flange spool into position.

Follow the approved gasket and joint procedure for tightening sequence, lubricants and target torque/tension. No generic torque chart is given here because bolt loading depends on joint design and materials.

After assembly, document any flange mismatch, interference or gasket rework under the site quality process.

FIELD CHECK If flange faces will not mate without force, correct the fit-up rather than overloading the bolts.

FIELD NOTES / TEST FEEDBACK

Concentric and eccentric reducers

Concentric reducers keep the large and small bore centerlines aligned. Eccentric reducers maintain a flat side and shift the smaller bore centerline relative to the larger bore.

Mark which side is to remain flat from the approved drawing and flow/service requirement: some runs must drain from the low point, while some pump suction arrangements aim to avoid vapor pockets at the high point.

Do not use a universal “flat on top” or “flat on bottom” rule. Elevation, flow direction, process fluid, vent/drain strategy and equipment nozzle geometry determine the correct arrangement.

When inserting a reducer on an ISO drawing, record both end sizes, orientation and its end-to-end takeoff. The app inline symbol does not replace a manufacturer dimension.

FIELD CHECK Inspect the eccentric side in the installed view; a plan-only sketch can hide the actual flat-side orientation.

FIELD NOTES / TEST FEEDBACK

Slope: fall over run

Slope is rise or fall divided by HORIZONTAL run. For a specified fall per foot q (in/ft), total fall (in) = q × horizontal run (ft). Do not multiply by the sloped pipe length.

Percent grade = 100 × rise/run using the SAME units. Angle = arctan(rise/run); sloping travel length = sqrt(run² + rise²).

Set a level datum and record upstream and downstream centerline elevations. Check all endpoints, vents, drains, high points and tie-ins before fabricating the spool.

Specified fall 1/8 in per foot across 24 ft horizontal.
Fall = 0.125 × 24 = 3.000 in.
Percent grade = 100 × (3/288) = 1.0417%.
Angle = arctan(3/288) ≈ 0.597°.
Pipe travel ≈ sqrt(288² + 3²) = 288.016 in.

FIELD CHECK Mark flow arrow and which end is HIGH. An undirected slope note is incomplete for layout.

FIELD NOTES / TEST FEEDBACK

Field-cut chain and spool length

Work from one approved centerline datum. List each component in order: elbows, tees, reducers, valves, flanges and straight pipe. Assign each its correct center-to-end or end-to-end takeoff.

A straight cut length is the specified datum-to-datum distance minus applicable fitting takeoffs and deliberate joint allowances. Do not subtract takeoffs twice from a dimension already given end-to-end.

Leave any bevel, root-gap, insertion, trim, shrink or field-fit allowance to the approved joint detail. A generic “weld shrinkage per joint” number is not reliable across pipe sizes and processes.

Illustrative center-to-center span = 72.000 in.
Fitting A center-to-end = 6.000 in; fitting B center-to-end = 4.500 in.
Nominal straight length = 72 - 6 - 4.5 = 61.500 in.
Apply project-specific end preparation and final cut requirements separately.

FIELD CHECK Record which dimensions are C-C, E-E, face-to-face or flange-face. Do not add allowances to an already defined finished piece.

FIELD NOTES / TEST FEEDBACK

Field-spool verification list

Before cutting: reconcile the latest ISO drawing, revision, line tag, NPS/OD, wall or schedule, grade, end preparation, fit-up allowance and datum. Walk the route and identify interferences and existing tie-ins.

Before welding or bolting: identify component markings, weld IDs, orientation, valve flow direction, bolt clocking, branch takeoffs, clearances, support points and achievable access. Preserve any project-required traceability.

Before handover: remeasure critical dimensions, flange face-to-face, centerline elevation, slope, branch orientation and valve access. Record approved field changes and close punch items in the job process.

Use signed design drawings, welding procedures, inspection plans and client specifications for all quality decisions. This checklist is an organizing aid, not an acceptance code.

FIELD CHECK Capture one clearly labeled datum photo or marked sketch with the measured value and revision.

FIELD NOTES / TEST FEEDBACK

Pressure-test preparation

Identify the approved test package, pressure boundary, temporary blinds, piping class, test medium, test pressure, hold criteria and responsible supervisor from the project procedure. This guide supplies no universal test pressure.

Confirm vents, drains, supports, calibrated instruments, isolation from equipment not included in the test, and provisions for safely filling, pressurizing and depressurizing.

Liquid testing may be preferable to pneumatic testing where permitted, but both require documented hazard controls. Never infer a safe pneumatic test or isolation scheme from a quick field calculation.

Verify valve positions and temporary parts against the approved test boundary. Check removal of test blinds and reinstatement after authorized closeout.

FIELD CHECK Test execution and acceptance must follow the governing code, permits and approved site procedure.

FIELD NOTES / TEST FEEDBACK

Everyday field math checks

Equal marks around pipe circumference: pitch = pi × actual OD / N. Angular increment = 360°/N. A wraparound reference line keeps the start mark consistent.

Distance between two planar offset points: T = sqrt($R^2 + V^2$). If an angle from horizontal is known, V = T sin(A) and R = T cos(A). Enter angles in degrees.

Pipe volume in gallons for a full straight round bore: Vgal = (pi × $ID^2/4$ × L)/231, with ID and L in inches. Do not use the volume estimate for pressure design or isolation decisions.

For a planar bolt circle, chord between adjacent holes = B sin(180°/N). All dimensions must use consistent units.

Illustrative ID = 4.000 in and length = 120 in.
V = pi × $4^2/4$ × 120 = 1507.965 cubic in.
Vgal = 1507.965 / 231 = 6.528 gal (approx).
This excludes elbows, branches, fittings, headspace and drainage residues.

FIELD CHECK If a number looks wrong, estimate magnitude, check units and verify the actual pipe size before reworking it.

FIELD NOTES / TEST FEEDBACK

REFERENCE

Cutting and pipe-size references

New edition 0.3 pages use selected factual values from clearly identified manufacturer cutting-tip charts. Their full current instructions govern all operating settings and limitations.

The nominal pipe/OD lookup lists common standard steel-pipe dimensions for field arithmetic. ASME publications and the actual material documentation control tolerances, schedule and purchased size.

This book provides reference calculations; it does not override engineered specifications, hot-work controls, rated lifting instructions, pressure-test plans or approved job procedures.

[15] Harris V1-101 cutting tip operating pressures, PN 9700378 (August 2023)
https://ch-delivery.lincolnelectric.com/api/public/content/bd29bc5b871a45aeb8ba5cf249542170?v=9a551e9d
[16] ESAB/Victor Type 101 acetylene CUTTING tips
https://esab.com/us/nam_en/products-solutions/product/gas-equipment/wear-parts/type-101-acetylene-cutting-tips/
[17] ASME B36.10 welded and seamless wrought steel pipe
https://www.asme.org/codes-standards/find-codes-standards/welded-and-seamless-wrought-steel-pipe

Straight lines with angle iron

1. Support and block the pipe so it cannot roll. Clean the marking area and contact surfaces. Check the angle against a known straight edge; reject bent, twisted or burred stock.

2. Establish the starting reference mark. If the line must represent installed top, find top first (page 66). An arbitrary straight line does not establish top or elevation.

3. Place the angle lengthwise with its open inside corner over the pipe. Both inside faces must contact the pipe along its length. Choose legs long enough to seat properly on the pipe diameter.

4. Rotate the angle around the pipe until one accessible marking edge passes through the reference mark. Keep both faces seated; do not twist one end to force alignment.

5. Hold or secure the angle without moving the pipe. Draw a thin line along the same edge. Keep the marker upright and use a material-compatible marker.

6. For a longer run, overlap the previous line at two separated points before extending it. Recheck against an independently established end mark.

FIELD CHECK Rocking, a raised weld seam or a dent can lift one contact face and skew the line. Reposition and verify; do not bridge an obstruction and assume accuracy.

Layout background: U.S. Navy Steelworker, Volume 2, Chapter 3; see page 69 for references.

FIELD NOTES / TEST FEEDBACK

Straight lines with strut

1. Use a straight, untwisted piece of channel strut. Remove nuts, brackets and debris. Inspect both lips for damage and check the proposed marking edge for straightness.

2. With the pipe blocked, place the open channel toward the pipe. Both longitudinal lips must bear evenly on the curved surface. A small pipe may enter the channel and touch the web instead: that strut size is unsuitable.

3. Check seating at both ends and the middle. Press lightly from side to side: rocking or a visible gap means you do not have a dependable reference.

4. Align one continuous marking edge with the established reference mark. Keep the channel parallel to the pipe axis with both lips seated. Do not use slot edges as a marking guide.

5. Mark along the same straight edge, then lift and reseat the strut to check repeatability. If the new position does not agree, resolve seating, straightness or pipe-shape problems before cutting.

6. Extend in overlapping lengths, matching the existing line at two separated points. Confirm the far end against a second reference mark.

FIELD CHECK Strut is a field straightedge, not a calibrated center finder. Use suitable angle iron or a purpose-made pipe layout tool if the lips cannot seat evenly. Neither tool establishes a square cut around the pipe.

FIELD NOTES / TEST FEEDBACK

Find top with a framing square

1. Use this method on a straight, round, horizontal pipe. Block it against rolling and check its axis with a level. For sloped pipe, use a suitable center finder or an approved survey method.

2. Check the square for squareness and the level by reversing it on the same reference surface. Clean the pipe and square. Use actual outside diameter (OD), not nominal pipe size.

3. Hold the square in a cross-section plane, perpendicular to the pipe axis. Put the inside horizontal edge against the crown and the inside vertical edge against one side. Both edges must be tangent to the pipe.

4. Level the horizontal leg while keeping both contacts seated. From the inside corner, measure OD / 2 along that horizontal leg toward the pipe center. Read from the true corner, not an offset scale zero.

5. Mark the pipe directly beneath that position. This is the 12 o’clock point for the present installed orientation. Repeat at another station without allowing the pipe to roll.

Example: actual OD = 4.500 in.
Half OD = 2.250 in = 2-1/4 in from the inside corner.
Using nominal 4 in would put the mark 1/4 in off.

FIELD CHECK Repeat from the opposite side: top marks should agree within the job tolerance. Recheck level, square position and roundness if they do not.

[18] U.S. Navy Steelworker, Volume 2, Figure 3-42 (reference on page 69).

Transfer and preserve top marks

1. Establish top independently at two separated cross sections using page 66. Mark each point and label the intended installed top before moving the pipe.

2. Connect those marks using the angle-iron or strut method. If the edge cannot pass through both marks while seated, investigate the pipe and the marks; do not force the tool.

3. To obtain side and bottom references, wrap a nonstretch strip squarely around the pipe, align its overlapping edges and mark one exact circumference. Remove it and divide that circumference into four equal parts.

4. Refit the strip with its zero at your top mark. Transfer quarter, half and three-quarter circumference marks: 3, 6 and 9 o’clock when viewed from your declared reference end. Carry those marks lengthwise as needed.

5. Add a viewing-direction arrow or name the reference end so another fitter cannot reverse left and right. Label the datum and spool identification outside the cut area.

6. If the pipe is rolled for welding, the mark still identifies intended installed top; it is no longer necessarily gravity top. Restore the documented orientation before checking slope and branch clocking.

FIELD CHECK Use a wraparound for a square circumferential line. A longitudinal straightedge and a circumferential wrap establish different references; check both before transferring a saddle or branch layout.

Choose reducer orientation

1. Confirm the fitting is eccentric, and check both sizes, material, wall/end preparation and the drawing revision. Its end centerlines are offset; a concentric reducer has a common centerline.

2. Read the specified clocking and elevation basis: flat on top (FOT), flat on bottom (FOB), or another orientation. If absent or conflicting, obtain the design clarification before fit-up.

3. For horizontal liquid pump suction, an eccentric reducer is commonly arranged flat on top to avoid an air pocket. Check the entire suction profile and the specific pump instructions; this is not a universal rule for every inlet arrangement.

4. Flat on bottom can preserve a specified bottom elevation. Pump inlet geometry, drainage and solids service require the approved design; do not choose orientation from a slogan.

5. Check flow direction, adjoining elbows, slope, vents/drains and the path air or liquid must take. Avoid creating an unintended local high or low pocket.

6. Identify the flat side with a straightedge along the reducer profile, allowing for weld bevels and end lands. Mark that side and the required installed top before assembly.

FIELD CHECK Flat side is not the same as a common centerline. Do not pull connected equipment into alignment to accommodate a wrongly clocked reducer.

[19] KSB, Centrifugal Pump Lexicon: Fitting. Reference link on page 69; equipment-specific installation instructions govern.

Fit and verify the reducer

1. Establish the installed top reference on both adjoining pipes. Support them at the specified elevations and slope. Transfer clocking marks to the reducer.

2. Dry-fit to the drawing. Check end-to-end length and the actual fitting offset from its dimensional data. For ideal matching external flat profiles, centerline offset is (large OD - small OD) / 2; verify the manufactured fitting.

3. Check the designated flat profile with a straightedge and the required slope with a level or survey. Flat on top describes clocking; it does not cancel a specified line slope. An external check does not prove internal drainage where walls differ.

4. Verify root gap, internal mismatch and end preparation against the WPS and project requirements. Check flange clocking, nozzle alignment and support contact without forcing the joint.

5. Tack only under the approved procedure, then recheck clocking, elevations, slope and joint alignment. Repeat critical checks after welding and before release.

FIELD CHECK Record reducer orientation on the spool sketch. Resolve unintended air or liquid pockets with the designer before final assembly.

[18] U.S. Navy Steelworker, Vol. 2, Figure 3-42 (public reprint)
https://constructionmanuals.tpub.com/14251/css/Quartering-the-Pipe-100.htm
[19] KSB, Fitting / reducers for avoiding air pockets
https://www.ksb.com/en-global/centrifugal-pump-lexicon/f

FIELD NOTES / TEST FEEDBACK